HW5 – Theory + SVM

1. <u>PAC Learning and VC dimension (30 pts)</u>

   Let $X = \mathbb{R}^2$. Let

$$C = H = \left\{ h(r_1, r_2) = \left\{ (x_1, x_2) \middle| \begin{matrix} x_1^2 + x_2^2 \geq r_1 \\ x_1^2 + x_2^2 \leq r_2 \end{matrix} \right\} \right\}, \text{ for } 0 \leq r_1 \leq r_2,$$

   the set of all origin-centered rings.

   a. (8 pts) What is the $VC(H)$? Prove your answer.

The VC dimension of a hypothesis class H is the largest number d such that there exists a set of d points that can be shattered by H. In this case H represents the set of points within a ring centered at the origin with an inner radius r_1 and outer radius r_2.

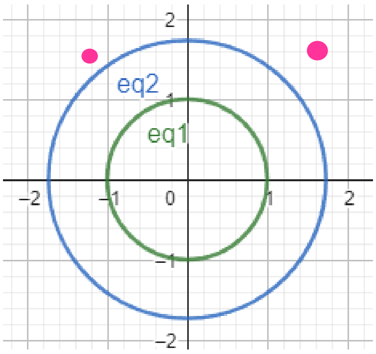
$VC(H) = 2$ means we have $2^2 = 4\ dichotomies$.

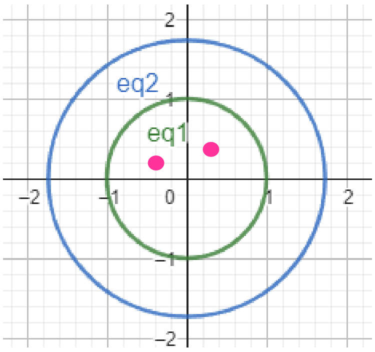
<u>Prove:</u>

Given $p_1 = (x_1, y_1)$ and $p_2 = (x_2, y_2)$, $\sqrt{x_1^2 + y_1^2} \neq \sqrt{x_2^2 + y_2^2}$ 2 different points and without loss of generality $\sqrt{x_1^2 + y_1^2} < \sqrt{x_2^2 + y_2^2}$.

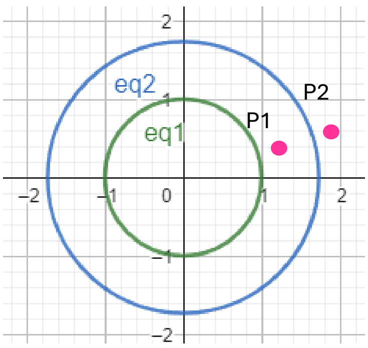
We can shatter the 4 dichotomies as follows:

$1.\ p_1 \to -, p_2 \to -: r_1 \leq r_2 < \sqrt{x_1^2 + y_1^2}$, or $\sqrt{x_2^2 + y_2^2} < r_1 \leq r_2$

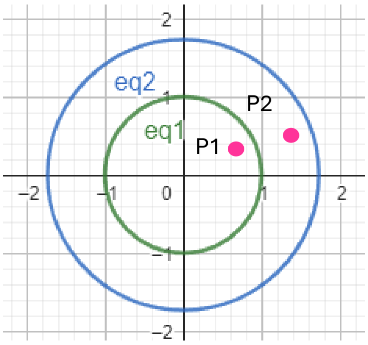




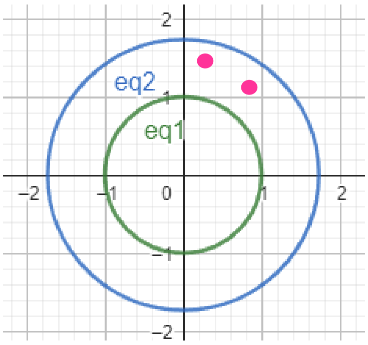
$2.\ p_1 \to +, p_2 \to -: r_1 \leq \sqrt{x_1^2 + y_1^2}, \sqrt{x_1^2 + y_1^2} \leq r_2 < \sqrt{x_2^2 + y_2^2}$



$3. p_1 \to -, p_2 \to +: \sqrt{x_1^2+y_1^2} < r_1 \le \sqrt{x_2^2+y_2^2}, \sqrt{x_2^2+y_2^2} \le r_2$



$4. p_1 \to +, p_2 \to +: r_1 \le \sqrt{x_1^2+y_1^2}, \sqrt{x_2^2+y_2^2} \le r_2$



To show that $VC(H) \neq 3$ we define $p_1 = (x_1, y_1), p_2 = (x_2, y_2)$ and $p_3 = (x_3, y_3)$.

without loss of generality $\sqrt{x_1^2+y_1^2} \le \sqrt{x_2^2+y_2^2} \le \sqrt{x_3^2+y_3^2}$.

Assume $p_1 \to -, p_2 \to +, p_3 \to -$. From the labeling of p_1 it implies that $r_1 > \sqrt{x_1^2+y_1^2}$.

From p_2 we get $r_1 \le \sqrt{x_2^2+y_2^2}$ and $r_2 \ge \sqrt{x_2^2+y_2^2}$. From p_3 we get the ring should end before p_3's distance so $r_2 < \sqrt{x_3^2+y_3^2}$.

Combining these, we get: $\sqrt{x_1^2+y_1^2} < r_1 \le \sqrt{x_2^2+y_2^2} \le r_2 < \sqrt{x_3^2+y_3^2}$ in contrast to the fact that p_1& p_3 are in the same group. Hence, the VC dimension cannot be 3 or more.

b. (14 pts) Describe a polynomial sample complexity algorithm L that learns C using H. State the time complexity and the sample complexity of your suggested algorithm. Prove all your steps.

In class we saw a bound on the sample complexity when H is finite.

$$m \ge \frac{1}{\varepsilon}\left(\ln|H| + \ln\frac{1}{\delta}\right)$$

When $|H|$ is infinite, we have a different bound:

$$m \geq \frac{1}{\varepsilon}\left(4 \log_2 \frac{2}{\delta} + 8VC(H) \log_2 \frac{13}{\varepsilon}\right)$$

The goal is to design an algorithm that identifies the smallest origin-centered ring that encloses all positive examples while excluding negative examples. This algorithm will operate efficiently with polynomial time complexity.

The Input: A dataset $\Delta = \{(x_i, y_i)\}_{i=1}^{m}$ consisting of m labeled points, where each $x_i \in R^2$ *and* $y_i \in \{+1, -1\}$.

The Output: A hypothesis $h \in H$ representing the smallest origin-centered ring that includes all positive points and excludes all negative points.

Steps:

1. Separate the dataset into two subsets:

   $\Delta^+ = \{(x_i, y_i) \mid y_i = +1\}$: *All points labeled as positive*.

   $\Delta^- = \{(x_i, y_i) \mid y_i = -1\})$: *All points labeled as negative*.

2. Determine the minimum and maximum distances from the origin for the positive points:

   - Calculate $r_{\min} = \min_{(x_i, y_i) \in \Delta^+} \sqrt{x_i^2 + y_i^2}$
   - Calculate $r_{\max} = \max_{(x_i, y_i) \in \Delta^+} \sqrt{x_i^2 + y_i^2}$

3. Adjust these radii to ensure no negative points fall within the defined ring:

   - For each $(x_j, y_j) \in \Delta^-$:
     - If $r_{\min} \leq \sqrt{x_j^2 + y_j^2} \leq r_{\max}$, update:
       - $r_{\min} = \min\left(r_{\min}, \sqrt{x_j^2 + y_j^2}\right)$
       - $r_{\max} = \max\left(r_{\max}, \sqrt{x_j^2 + y_j^2}\right)$

4. Return the hypothesis $L(\Delta) = h(r_{\min}, r_{\max})$, which defines the ring from $r_{\min}$ *to* $r_{\max}$

Time Complexity:

- Calculating $r_{\min}$ *and* $r_{\max}$ by iterating over all positive points takes $O(m)$.
- Adjusting the radii based on the negative points also takes $O(m)$.
- Therefore, the overall time complexity of the algorithm is $O(m)$, where m is the number of samples.

Sample Complexity:

Given $\varepsilon > 0$ and $\delta > 0$, our objective is to ensure that the hypothesis h generated by the learning algorithm accurately approximates the true concept c within an error margin ε. To achieve this, we define ε-coverage consisting of two sets:

1. Inner set $S_1(\varepsilon)$ –
$$S_1(\varepsilon) = \{(x, y) \mid r_1 \le x_i^2 + y_i^2 \le r_1 + \alpha\}$$
This set contains points near the inner boundary of the ring, helping to ensure that the inner radius is not underestimated.
2. Outer set- $S_2(\varepsilon)$-
$$S_2(\varepsilon) - \{(x, y) \mid r_2 - \beta \le x_i^2 + y_i^2 \le r_2\}$$
This set includes points near the outer boundary, ensuring that the outer radius is not overestimated.

The parameters α and β are selected such that the probability of a point falling within $\pi(S_1(\varepsilon))$ and $\pi(S_2(\varepsilon))$ is $\frac{\varepsilon}{2}$. This ensures that the regions are small enough to fall within the error margin ε but substantial enough to be measurable.

__**Define the interval:**__ $I(\varepsilon) = h(r_1 + a, r_2 - \beta)$ This interval is a ring with an inner radius of $r_1 + a$ and an outer radius of $r_2 - \beta$.

__**Probability Calculation:**__

1. Suppose the error of the hypothesis produced by our learning algorithm L represented as $L(\Delta_m(c))$ is greater than ε:
$$err_{\Pi}(L(\Delta_m(c)), C) > \varepsilon$$
This indicates that the hypothesis $L(\Delta_m(c))$ does not sufficiently approximate the true concept c within the specified error margin ε.
2. If the error condition holds, it means the training set $\Delta_m(c)$ did not cover the sets $S_1(\varepsilon)$ or $S_2(\varepsilon)$. Essentially, no points from the training set fell within these sets.
3. We need to determine the probability that none of the m points in the training set fall into $S_1(\varepsilon)$ or $S_2(\varepsilon)$.

   - The probability that none of the m points fall into $(S_1(\varepsilon))$ is:
   $$\pi_m(\Delta_m(c) \cap S_{1(\varepsilon)} = \emptyset) = \left(1 - \Pi(S_1(\varepsilon))\right)^m = \left(1 - \frac{\varepsilon}{2}\right)^m$$
   - Similarity ,for The probability that none of the m points fall into $(S_2(\varepsilon))$ is:
   $$\pi_m(\Delta_m(c) \cap S_{2(\varepsilon)} = \emptyset) = \left(1 - \Pi(S_2(\varepsilon))\right)^m = \left(1 - \frac{\varepsilon}{2}\right)^m$$

4. The combined probability that the training set misses either $S_1(\varepsilon)$ or $S_2(\varepsilon)$.

$$\pi_m\left(err_{\pi}(L(\Delta_m(c)), c) > \varepsilon\right) \le \left(\pi_m(\Delta_m(c) \cap S_{1(\varepsilon)} = \emptyset) + \pi_m(\Delta_m(c) \cap S_{2(\varepsilon)} = \emptyset)\right) \le 2\left(1 - \frac{\varepsilon}{2}\right)^m$$

5. Ensuring Confidence: $2\left(1-\frac{\varepsilon}{2}\right)^m \le \delta$
6. Solving for m:

$$\ln\left(2\left(1-\frac{\varepsilon}{2}\right)^m\right) \le \ln(\delta)$$

$$\ln(2) + m\, ln\left(1-\frac{\varepsilon}{2}\right) \le \ln(\delta)$$

Using $\ln(1-x) \approx -x$ for small x:

$$\ln(2) - m\frac{\varepsilon}{2} \le \ln(\delta)$$

$$m\frac{\varepsilon}{2} \ge \ln(2) - \ln(\delta)$$

$$m \ge \frac{2\ln(2) + 2\ln\left(\frac{1}{\delta}\right)}{\varepsilon}$$

Therefore, the sample complexity is:

$$m(\varepsilon,\delta) \ge \frac{2}{\varepsilon}\left(\ln(2) + \ln\left(\frac{1}{\delta}\right)\right)$$

This ensures the hypothesis h generated by the learning algorithm is within the desired accuracy ε with confidence $1-\delta$.

Conclusion

This algorithm efficiently learns the hypothesis class H with polynomial sample complexity and time complexity, making it a robust solution for identifying the smallest origin-centered ring encompassing all positive points while excluding negative points.

c. (8 pts) You want to get with 95% confidence a hypothesis with at most 5% error. Calculate the sample complexity with the bound that you found in b and the above bound for infinite $|H|$. In which one did you get a smaller m? Explain.

To find the sample complexity with 95% confidence (meaning $\delta = 0.05$) and an error margin of at most 5% ($\varepsilon = 0.05$), we will use both the bounds derived in part (b) and the standard bound for infinite hypothesis classes.

<u>Using the Bound Derived in Part (b):</u>

$$m(\varepsilon = 0.05, \delta = 0.05) \ge \frac{2}{0.05}\left(\ln(2) + \ln\left(\frac{1}{0.05}\right)\right)$$
$$m \ge 40\,(\ln(2) + \ln(20))$$

$$m \ge 40\,(\ln(2 * 20)$$
$$m \ge 40\,(\ln(40))$$

$$m \geq 147.56$$

Using the Standard Bound for Infinite Hypothesis Classes:

$$m \geq \frac{1}{\varepsilon}\left(4\log_2\frac{2}{\delta} + 8VC(H)\log_2\frac{13}{\varepsilon}\right)$$

From part (a), $VC(H)$=2.

$$m \geq \frac{1}{0.05}\left(4\log_2\frac{2}{0.05} + 8 * 2\log_2\frac{13}{0.05}\right)$$

$$m \geq 20(4\log_2 40 + 16\log_2 260)$$

$$m \geq 2992.96$$

Comparing the two bounds, we see that the sample complexity derived using the bound from part (b) is significantly smaller ($m \geq 147.56$) compared to the standard bound for infinite hypothesis classes ($m \geq 2992.96$). This difference is because that the bound derived in part (b) leverages the specific geometric properties of the hypothesis class, leading to a tighter and more efficient bound.

2. VC dimension (20 pts)

Let $X = \mathbb{R}$ and $n \in \mathbb{N}$.

Define "x-node decision tree" for any $x = 2^n - 1$ to be a full binary decision tree with x nodes (including the leaves).

Let H_m be the hypothesis space of all "x-node decision tree" with $n \leq m$.

a. (5 pts) What is the $VC(H_3)$? Prove your answer.

H_3 includes all possible full binary trees with up to 3 nodes.
Here are the configurations possible:

- **$x = 1$: Only the root node.**
  A tree with 1 node (just the root) can classify a single point. It can decide to classify it as positive or negative.
  Hence, H_3 can shatter 1 point.
- **$x = 2$: Root node with one child.**
  With 2 nodes (root and one child), H_3 can classify two points.
  For instance, one point could be in the left subtree and the other in the right subtree, or both points could be in the same subtree but classified differently by the leaf nodes.
  Thus, H_3 can shatter 2 points.
- **$x = 3$: Root node with two children (2 leaves).**
  With 3 nodes (root and two children), H_3 cannot shatter all possible configurations of 3 points. There aren't enough nodes to create distinct classifications for all subsets of 3 points. Specifically, there aren't enough leaf nodes to represent all possible dichotomies of 3 points. Therefore, H_3 cannot shatter 3 points.

Therefore, the correct VC dimension for H_3 is 2.

b. (15 pts) What is the $VC(H_m)$? Prove your answer.

For $n \leq m$, the largest n would be m. Hence, the largest x is $2^{\mathrm{m}} - 1$ nodes.

The hypothesis space H_m includes all trees with up to $2^m - 1$ nodes, i.e., full binary trees of height up to $m - 1$ (since a tree of height $m - 1$ has $2^m - 1$ nodes).

A full binary tree of height $m - 1$ has $2^m - 1$ nodes. Typically, a binary decision tree with d leaves can shatter up to d points. For our largest tree, the number of leaves is 2^{m-1}.

The VC dimension of a decision tree with $2^m - 1$ nodes (which corresponds to a depth of $m - 1$) is 2^{m-1}. This is because it can create 2^{m-1} decision boundaries (splits) to separate up to 2^{m-1} points in any combination.

Thus, $VC(H_m) = 2^{m-1}$.

3. Kernels and mapping functions (25 pts)
   a. (20 pts) Let $K(x,y) = (x \cdot y + 1)^3$ be a function over $\mathbb{R}^2 \times \mathbb{R}^2$ (i.e., $x, y \in \mathbb{R}^2$).

      Find ψ for which K is a kernel. (It may help to first expand the above term on the right-hand side).

Solution:

Expand the Kernel Function: $K(x,y) = (x_1y_1 + x_2y_2 + 1)^3$

Expand this expression using the binomial theorem:

$$(a+b+c)^3 = a^3 + b^3 + c^3 + 3a^2b + 3a^2c + 3ab^2 + 3b^2c + 3ac^2 + 3bc^2 + 6abc$$

Here $a = x_1y_1, b = x_2y_2, c = 1$

$$(x_1y_1 + x_2y_2 + 1)^3$$
$$= (x_1y_1)^3 + (x_2y_2)^3 + 1^3 + 3(x_1y_1)^2(x_2y_2) + 3(x_1y_1)^2 1 + 3(x_1y_1)(x_2y_2)^2 + 3(x_2y_2)^2 1 + 3(x_1y_1)1^2 + 3(x_2y_2)1^2 + 6(x_1y_1)(x_2y_2)1$$
$$= x_1^3y_1^3 + x_2^3y_2^3 + 1 + 3x_1^2y_1^2x_2y_2 + 3x_1^2y_1^2 + 3x_1y_1x_2^2y_2^2 + 3x_2^2y_2^2 + 3x_1y_1 + 3x_2y_2 + 6x_1y_1x_2y_2$$

Rearranging for better visualization:

$$K(x,y) = (x_1y_1 + x_2y_2 + 1)^3 =$$
$$= x_1^3y_1^3 + 3x_1^2y_1^2x_2y_2 + 3x_1y_1x_2^2y_2^2 + x_2^3y_2^3 + 3x_1^2y_1^2 + 6x_1y_1x_2y_2 + 3x_2^2y_2^2 + 3x_1y_1 + 3x_2y_2 + 1$$

Identifying the Feature Mapping $\psi(x)$

We can represent the kernel as an inner product in a higher-dimensional space as follows:

$$K(x,y) = \psi(x) \cdot \psi(y)$$

Reconstructing the final expression as vectors:

$$K(x,y) = x_1^3y_1^3 + 3x_1^2y_1^2x_2y_2 + 3x_1y_1x_2^2y_2^2 + x_2^3y_2^3 + 3x_1^2y_1^2 + 6x_1y_1x_2y_2 + 3x_2^2y_2^2 + 3x_1y_1 + 3x_2y_2 + 1$$
$$= \left(x_1^3, \sqrt{3}x_1^2x_2, \sqrt{3}x_1x_2^2, x_2^3, \sqrt{3}x_1^2, \sqrt{6}x_1x_2, \sqrt{3}x_2^2, \sqrt{3}x_1, \sqrt{3}x_2, 1\right) \cdot$$
$$\left(y_1^3, \sqrt{3}y_1^2y_2, \sqrt{3}y_1y_2^2, y_2^3, \sqrt{3}y_1^2, \sqrt{6}y_1y_2, \sqrt{3}y_2^2, \sqrt{3}y_1, \sqrt{3}y_2, 1\right) = \psi(x) \cdot \psi(y)$$

Thus, the mapping function $\psi(x)$ that maps x from $\mathbb{R}^2$ to a higher-dimensional space $\mathbb{R}^{10}$ is:

$$\boxed{\psi(x) = \left(x_1^3, \sqrt{3}x_1^2x_2, \sqrt{3}x_1x_2^2, x_2^3, \sqrt{3}x_1^2, \sqrt{6}x_1x_2, \sqrt{3}x_2^2, \sqrt{3}x_1, \sqrt{3}x_2, 1\right)}$$

   b. (2 pts) What did we call the function ψ in class if we remove all coefficients?

If we remove all the coefficients from the function ψ

$$\psi(x) = (x_1^3, x_2^3, x_1^2x_2, x_1x_2^2, x_1^2, x_2^2, x_1x_2, x_1, x_2, 1)$$

We get the rational variety of degree 3.

c. (3 pts) How many multiplication operations do we save by using $K(x,y)$ versus $\psi(x)\cdot\ \psi(y)$?

To compute $\psi(x)\cdot\psi(y)$, we calculate:

$\psi(x)\cdot\psi(y) = (x_1^3)(y_1^3) + \left(\sqrt{3}x_1^2x_2\right)(\sqrt{3}y_1^2y_2) + (\sqrt{3}x_1x_2^2)(\sqrt{3}y_1y_2^2) + (x_2^3)(y_2^3) + (\sqrt{3}x_1^2)(\sqrt{3}y_1^2) + (\sqrt{6}x_1x_2)(\sqrt{6}y_1y_2) + (\sqrt{3}x_2^2)(\sqrt{3}y_2^2) + (\sqrt{3}x_1)(\sqrt{3}y_1) + (\sqrt{3}x_2)(\sqrt{3}y_2) + (1)(1).$

a. $(1)(1)$ – **1 multiplication.**
b. $(\sqrt{3}x_2)(\sqrt{3}y_2)$ – **3 multiplications:**
   1. $\sqrt{3}\cdot x_2 = \sqrt{3}x_2$
   2. $\sqrt{3}\cdot y_2 = \sqrt{3}y_2$
   3. $\sqrt{3}x_2\cdot\sqrt{3}y_2$
c. $(\sqrt{3}x_1)(\sqrt{3}y_1)$ – **3 multiplications:**
   1. $\sqrt{3}\cdot x_1 = \sqrt{3}x_1$
   2. $\sqrt{3}\cdot y_1 = \sqrt{3}y_1$
   3. $\sqrt{3}x_1\cdot\sqrt{3}y_1$
d. $(\sqrt{3}x_2^2)(\sqrt{3}y_2^2)$ – **5 multiplications:**
   1. $x_2\cdot x_2 = x_2^2$
   2. $\sqrt{3}\cdot x_2^2$
   3. $y_2\cdot y_2 = y_2^2$
   4. $\sqrt{3}\cdot y_2^2$
   5. $\sqrt{3}x_2^2\cdot\sqrt{3}y_2^2$
e. $\sqrt{6}x_1x_2)(\sqrt{6}y_1y_2)$ – **5 multiplications:**
   1. $x_1\cdot x_2 = x_1x_2$
   2. $\sqrt{6}\cdot x_1x_2$
   3. $y_1\cdot y_2 = y_1y_2$
   4. $\sqrt{6}\cdot y_1y_2$
   5. $\sqrt{6}x_1x_2\cdot\sqrt{6}y_1y_2$
f. $(\sqrt{3}x_1^2)(\sqrt{3}y_1^2)$ – **5 multiplications:**
   1. $x_1\cdot x_1 = x_1^2$
   2. $\sqrt{3}\cdot x_1^2$
   3. $y_1\cdot y_1 = y_1^2$
   4. $\sqrt{3}\cdot y_1^2$
   5. $\sqrt{3}x_1^2\cdot\sqrt{3}y_1^2$
g. $(x_2^3)(y_2^3)$ – **5 multiplications:**
   1. $x_2\cdot x_2\cdot x_2$ 2 multiplications.
   2. $y_2\cdot y_2\cdot y_2$ 2 multiplications.
   3. $x_2^3\cdot y_2^3$

h. $(\sqrt{3}x_1x_2^2)(\sqrt{3}y_1y_2^2)$ – **7 multiplications**:
   1. $x_1 \cdot x_2 \cdot x_2 = x_1x_2^2$ 2 multiplications.
   2. $\sqrt{3} \cdot x_1x_2^2$
   3. $y_1 \cdot y_2 \cdot y_2 = y_1y_2^2$ 2 multiplications.
   4. $\sqrt{3} \cdot y_1y_2^2$
   5. $\sqrt{3}x_1x_2^2 \cdot \sqrt{3}y_1y_2^2$

i. $(\sqrt{3}x_1^2x_2)(\sqrt{3}y_1^2y_2)$ – **7 multiplications**:
   1. $x_1 \cdot x_1 \cdot x_2 = x_1^2x_2$ 2 multiplications.
   2. $\sqrt{3} \cdot x_1^2x_2$
   3. $y_1 \cdot y_1 \cdot y_2 = y_1^2y_2$ 2 multiplications.
   4. $\sqrt{3} \cdot y_1^2y_2$
   5. $\sqrt{3}x_1^2x_2 \cdot \sqrt{3}y_1^2y_2$

j. $(x_1^3)(y_1^3)$ – **5 multiplications**:
   1. $x_1 \cdot x_1 \cdot x_1$ 2 multiplications.
   2. $y_1 \cdot y_1 \cdot y_1$ 2 multiplications.
   3. $x_1^3 \cdot y_1^3$

Total multiplications: $5 + 7 + 7 + 5 + 5 + 5 + 5 + 3 + 3 + 1 = 46$.

To compute $K(x, y) = (x \cdot y + 1)^3$:
1. $x \cdot y = x_1y_1 + x_2y_2$ **2 multiplications**.
2. $(\alpha)^3 = \alpha \cdot \alpha \cdot \alpha$ **2 multiplications**.

Total multiplications: $2 + 2 = 4$.

Therefore, we have saved a total of $46 - 4 = 42$.

4. Lagrange multipliers (15 pts)

Let $f(x,y) = 2x - y$. Find the minimum and the maximum points for f under the constraint $g(x,y) = \frac{x^2}{4} + y^2 = 1$.

To find the minimum and maximum points of the function $f(x,y) = 2x - y$ subject to the constraint $g(x,y) = \frac{x^2}{4} + y^2 = 1$ we can use the method of Lagrange multipliers. This involves setting up the following equations:

1. $\nabla f = \lambda \nabla g$
2. The constraint equation $g(x,y) = 1$

Solution:

1. Compute the gradients:

$$\nabla \mathrm{f} = \left(\frac{\partial \mathrm{f}}{\partial \mathrm{x}}, \frac{\partial \mathrm{f}}{\partial \mathrm{y}}\right) = (2,-1) \qquad \nabla \mathrm{g} = \left(\frac{\partial \mathrm{g}}{\partial \mathrm{x}}, \frac{\partial \mathrm{g}}{\partial \mathrm{y}}\right) = \left(\frac{x}{2}, 2y\right)$$

2. Set up the equations with the Lagrange multiplier $\nabla f = \lambda \nabla g$:
   This gives us the system of equations:
   $$2 = \lambda \frac{x}{2} \Longrightarrow x = \frac{4}{\lambda} \Longrightarrow \lambda = \frac{4}{x}$$
   $$-1 = \lambda 2y \Longrightarrow \lambda = -\frac{1}{2y}.$$
3. Set the two expressions for λ equal to each other:
   $$\frac{4}{x} = -\frac{1}{2y} \Longrightarrow 8y = x \Longrightarrow x = -8y$$
4. Substitute $x = -8y$ into the constraint:
   $$g(x,y) = \frac{x^2}{4} + y^2 = \frac{(-8y)^2}{4} + y^2 = 16y^2 + y^2 = 1 \Longrightarrow 17y^2 = 1$$
   $$\Longrightarrow y_{1,2} = \pm\frac{1}{\sqrt{17}}$$
   Substituting back for x:
   $$x = -8y \Longrightarrow x = -8\left(\pm\frac{1}{\sqrt{17}}\right) \Longrightarrow x_{1,2} = \pm\frac{8}{\sqrt{17}}$$
5. Evaluate $f(x,y)$ at the critical points:

$$f\left(\frac{8}{\sqrt{17}}, -\frac{1}{\sqrt{17}}\right) = 2\frac{8}{\sqrt{17}} - \left(-\frac{1}{\sqrt{17}}\right) = \frac{17}{\sqrt{17}} = \sqrt{17}$$
$$f\left(-\frac{8}{\sqrt{17}}, \frac{1}{\sqrt{17}}\right) = 2\left(-\frac{8}{\sqrt{17}}\right) - \frac{1}{\sqrt{17}} = -\frac{17}{\sqrt{17}} = -\sqrt{17}$$

Conclusion:

The maximum value of $f(x, y)$ under the given constraint is $\sqrt{17}$ at the point $\left(\frac{8}{\sqrt{17}}, -\frac{1}{\sqrt{17}}\right)$.

The minimum value of $f(x, y)$ under the given constraint is $-\sqrt{17}$ at the point $\left(-\frac{8}{\sqrt{17}}, \frac{1}{\sqrt{17}}\right)$.